INTRODUCTION TO AIRCRAFT STRUCTURES-MID TERM EXAM SOLUTION

Q1. The slope of the bending moment diagram at any section of a beam is equal to the (0.5)

1. **Shear force at that section
2. Load intensity at that section
3. Bending moment at that section
4. Deflection at that section

Q2. The principal stresses are (0.5)

1. ** The maximum and minimum normal stresses on a plane where shear stress is zero
2. The maximum shear stresses
3. The average normal stresses
4. Always equal

Q3. The modulus of elasticity of a material: (0.5)

1. Increases with strain
2. ** Decreases with strain beyond elastic limit
3. Remains constant throughout
4. Dependent on strain rate only

Q4. The principal stresses of a two-dimensional stress element are (0.5)

1. Normal stresses acting on the max. shear stress planes
2. Shear stresses acting on the principal planes
3. ** Normal stresses acting on the principal planes
4. Normal stresses acting on the max. shear stress planes

Q5. A cantilever beam 3 m long carries a uniformly distributed load of 4 kN/m. What is the bending moment at the fixed end? (0.5)

1. 36 kNm
2. 24 kNm
3. ** 18 kNm
4. 12 kNm

Q6. For a simply supported beam with a point load P located at the midpoint, the shear force diagram (SFD) will (0.5)

1. ** Show a sudden jump at the location of the point load.
2. Show a triangular distribution with maximum shear at the supports
3. Be a constant line along the length of the beam.
4. Be parabolic with zero shear at the midpoint.

Q7. The bending stress distribution in a beam is: (0.5)

1. Parabolic with maximum at neutral axis
2. ** Linear with zero stress at neutral axis and maximum at outer fibers
3. Uniform over the cross section

4. Randomly varying

Q8. The moment of resistance of a beam section is: (0.5)

1. Product of bending moment and section modulus
2. ** Product of maximum bending stress and section modulus
3. Ratio of bending moment to section modulus
4. Ratio of bending stress to section modulus

Q9. The angle between physical plane and principal plane is half of the angle subtended at the center by points on Mohr's Circle. If angle subtended is 60°, the physical plane angle is: (0.5)

1. 60°
2. ** 30°
3. 120°
4. 90°

Q10. Which of the following is true for principal stresses from Mohr's Circle? (0.5)

1. ** Principal stresses lie on the horizontal axis of Mohr's Circle
2. Sum of principal stresses equals twice the x-normal stress
3. Principal stresses equal the shear stresses
4. Principal stresses are always negative

Q11. In what way are shear force and bending moment diagrams useful for structural design? (2)

Shear force and bending moment diagrams are essential tools in structural design because they help engineers identify maximum internal stresses and determine safe sizing and material selection for beams and other structural elements.

Main Advantages

- Shear force and bending moment diagrams allow engineers to locate the points of maximum shear and bending stress, which are critical for ensuring that structural members can safely support applied loads without failure.
- These diagrams enable precise calculation and visualization of stress distribution along a beam, guiding decisions for cross-sectional dimensions and material choices in design

Q12. What is pure bending? Describe the shear force and bending moment diagrams for a beam segment in pure bending. (2)

Pure bending refers to the condition in which a beam is subjected only to a constant bending moment, without any shear force present along the length. This occurs when equal and opposite couples are applied at the ends, causing the beam to bend into an arc of a circle, with the top fibers in compression and bottom fibers in tension.

Shear Force and Bending Moment Diagrams in Pure Bending

- For a beam segment in pure bending, the shear force diagram will show zero shear force throughout the segment.
- The bending moment diagram will be a horizontal line, indicating a constant moment along the length where pure bending occurs.

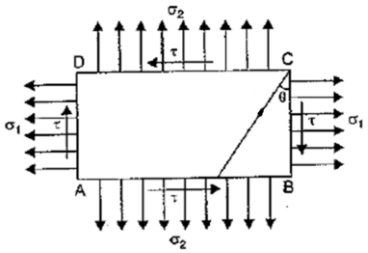
Q13. Explain briefly the different types of structural supports commonly used in beams, highlighting their representations, reaction forces, and the types of loads or movements they resist. (3)

S.no	Types of Support	Representation by	Reaction Force	Resisting Load
1.	**Roller Support**		Vertical	Vertical loads
2.	**Pinned Support**		Horizontal and vertical	Vertical and horizontal loads
3.	**Fixed Support**		Horizontal, vertical and moments	All types of loads Horizontal, vertical and Moments
4.	**Simple Support**		Vertical	Vertical loads

Q14. Explain briefly the step-by-step procedure to construct Mohr's Circle for a given state of plane stress, mentioning how the normal and shear stresses are plotted and how the principal stresses and maximum shear stresses are identified from the circle. (3)

- Sign Conventions:

1. Tensile normal stress is taken as +ve.
2. Compressive normal stress is taken as –ve.
3. Shearing stress producing clockwise moment in the element is -ve (on the major plane).



Construction of Mohr's circle:

1. Select a cartesian co-ordinate system with origin O and x-axis as direct stress axis and y-axis as shearing stress axis.
2. Locate point E (σ_1,τ) representing state of stress on the plane BC.
3. Locate point F (σ_2,τ) representing state of stress on the plane CD.
4. Join EF, let the point of intersection with the direct stress axis be G.
5. With G as center and GF=GE as radius draw a circle. This is Mohr's Circle.
6. Draw a line GJ making an angle 2Θ with GE.
7. From J, draw JK perpendicular to x-axis.
8. Join OJ, the length of OJ represents the resultant stress on the oblique plane GJ.
9. Lengths OK and JK represents the normal stress and tangential stress respectively.

Note:

1. Θ (Analytical) ⇒ 2Θ (Mohr's Circle)

Q15. Determine the maximum stress induced in a cast iron pipe of external diameter 40 mm, of internal diameter 20 mm and of length 4 meteres when the pipe is supported at its ends and carries a point load of 80 N at its centre. (3)

Sol. Given :

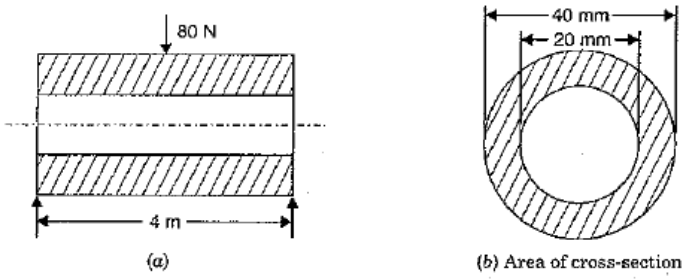
External dia., $D = 40$ mm

Internal dia., $d = 20$ mm

Length, $L = 4 \text{ m} = 4 \times 1000 = 4000$ mm

Point load, $W = 80$ N

In case of simply supported beam carrying a point load at the centre, the maximum bending moment is at the centre of the beam.



(a) (b) Area of cross-section

Fig. 7.6

And maximum B.M. $= \dfrac{W \times L}{4}$

$\therefore$ Maximum B.M. $= \dfrac{80 \times 4000}{4} = 8 \times 10^4$ Nmm

$\therefore$ $M = 8 \times 10^4$ Nmm

Fig. 7.6 (b) shows the cross-section of the pipe.

Moment of inertia of hollow pipe,

$$I = \frac{\pi}{64}[D^4 - d^4]$$

$$= \frac{\pi}{64}[40^4 - 20^4] = \frac{\pi}{64}[2560000 - 160000]$$

$$= 117809.7 \text{ mm}^4$$

Now using equation (7.4),

$$\frac{M}{I} = \frac{\sigma}{y} \qquad \ldots(i)$$

when y is maximum, stress will be maximum. But y is maximum at the top layer from the N.A.

$\therefore$ $y_{max} = \dfrac{D}{2} = \dfrac{40}{2} = 20$ mm

The above equation (i) can be written as

$$\frac{M}{I} = \frac{\sigma_{max}}{y_{max}}$$

$\therefore$ $\sigma_{max} = \dfrac{M}{I} \times y_{max}$

$$= \frac{8 \times 10^4 \times 20}{117809.7} = \mathbf{13.58 \text{ N/mm}^2}. \quad \textbf{Ans.}$$

Q16. Determine the modulus of rigidity and the bulk modulus of a cylindrical bar of diameter 30 mm and length of 1.5 m if the longitudinal strain in a bar during a tensile stress is four times lateral strain. Also find the change in volume when the bar is subjected to a hydrostatic pressure of 100 N/mm^2. Take E=1×10^5 N/mm^2. (4)

Sol. Given :

Dia. of bar, $d = 30$ mm

Length of bar, $L = 1.5 \text{ m} = 1.5 \times 1000 = 1500$ mm

$\therefore$ Volume of bar, $V = \frac{\pi}{4} d^2 \times L = \frac{\pi}{4} \times 30 \times 1500$

$= 1060287.52 \text{ mm}^3$

Longitudinal strain $= 4 \times$ Lateral strain

Hydrostatic pressure, $p = 100 \text{ N/mm}^2$

$\therefore \quad \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{1}{4} = 0.25$

or Poisson's ratio, $\mu = 0.25$

Let C = Modulus of rigidity

K = Bulk modulus

E = Young's modulus $= 1 \times 10^5 \text{ N/mm}^2$

Using equation (2.16), we get

$$E = 2C(1 + \mu)$$

or $$1 \times 10^5 = 2C(1 + 0.25)$$

$\therefore \quad C = \frac{1 \times 10^5}{2 \times 1.25} = \mathbf{4 \times 10^4 \text{ N/mm}^2}$. **Ans.**

For bulk modulus, using equation (2.11), we get

$$E = 3K(1 - 2\mu)$$

or $$1 \times 10^5 = 3K(1 - 2 \times 0.25)$$

$\therefore \quad K = \frac{1 \times 10^5}{3 \times 0.5} = \mathbf{0.667 \times 10^5 \text{ N/mm}^2}$. **Ans.**

Now using equation (2.9), we get

$$K = \frac{p}{\text{Volumetric strain}} = \frac{p}{\left(\frac{dV}{V}\right)}$$

where $p = 100 \text{ N/mm}^2$

$\therefore \quad 0.667 \times 10^5 = \frac{100}{\left(\frac{dV}{V}\right)}$

or $$\frac{dV}{V} = \frac{100}{0.667 \times 10^5} = 1.5 \times 10^{-3}$$

$\therefore \quad dV = V \times 1.5 \times 10^{-3} = 1060287.52 \times 1.5 \times 10^{-3}$

$= \mathbf{1590.43 \text{ mm}^3}$. **Ans.**

Q17. Derive the general expressions for shear force and bending moment at any section of a simply supported beam subjected to a uniformly varying load, which starts from zero intensity at the left support and increases linearly to *w* per unit length at the right support. Also, draw the shape of the corresponding shear force and bending moment diagrams. (4)

Taking moments about A, we get

$$R_B \times L = \left(\frac{w.L}{2}\right).\frac{2}{3}L \qquad \left[\text{Total load}\left(=\frac{w.L}{2}\right)\text{is acting }\frac{2}{3}L\text{ from }A\right]$$

$$\therefore \qquad R_B = \frac{w.L}{3}$$

and $$R_A = \text{Total load on beam} - R_B = \frac{w.L}{2} - \frac{w.L}{3} = \frac{w.L}{6}$$

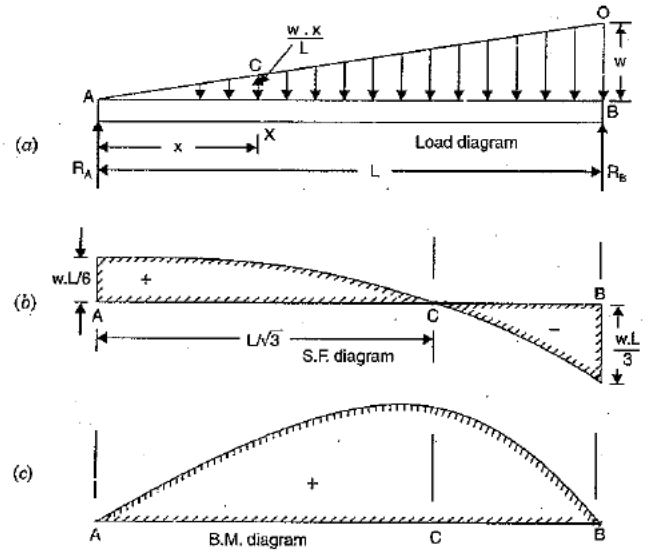
Consider any section X at a distance x from end A. The shear force at X is given by,

$$F_x = R_A - \text{load on length } AX = \frac{w.L}{6} - \frac{w.x}{L}.\frac{x}{2}$$

$$\left(\text{Load on } AX = \frac{AX\,.\,CX}{2} = \frac{x\,.\,w\,.\,x}{2\,.\,L}\right)$$

$$= \frac{wL}{6} - \frac{wx^2}{2L} \qquad \ldots(i)$$

Equation (i) shows that S.F. varies according to parabolic law.



At A, $x = 0$ hence, $\quad F_A = \frac{w.L}{6} - \frac{w}{2L} \times 0 = \frac{w.L}{6}$

At B, $x = L$, hence, $\quad F_B = \frac{w.L}{6} - \frac{w.L^2}{2L} = \frac{w.L}{6} - \frac{w.L}{2} = \frac{w.L - 3w.L}{6} = -\frac{2w.L}{6} = -\frac{w.L}{3}$

The shear force is $+\frac{w.L}{6}$ at A and it decreases to $-\frac{w.L}{3}$ at B according to parabolic law.

Somewhere between A and B, the S.F. must be zero. Let the S.F. be zero to a distance x from A. Equating the S.F. to zero in equation (i), we get

$$0 = \frac{wL}{6} - \frac{wx^2}{2L} \qquad \text{or} \qquad \frac{wx^2}{2L} = \frac{w.L}{6}$$

or $$x^2 = \frac{w.L}{6} \times \frac{2L}{w} = \frac{L^2}{3}$$

$$\therefore \qquad x = \frac{L}{\sqrt{3}} = 0.577\,L$$

B.M. Diagram

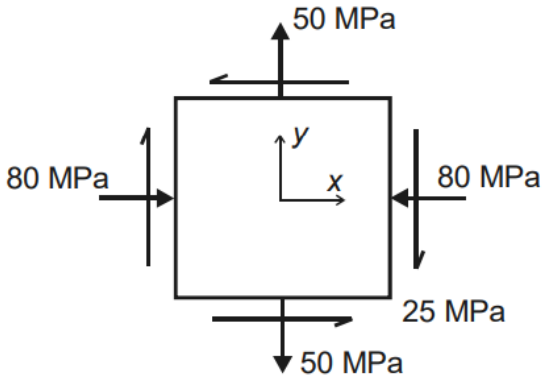
The B.M. is zero at A and B.

The B.M. at the section X at a distance x from the end A is given by,

$$M_x = R_A x - \text{Load on length } AX\,.\,\frac{x}{3} \qquad \left(\because \text{ Load on } AX \text{ is acting at } \frac{x}{3} \text{ from } X\right)$$

$$= \frac{w.L}{6}.x - \frac{wx^2}{2L}.\frac{x}{3} = \frac{wL}{6}.x - \frac{wx^3}{6L}$$

Q18. The state of plane stress at a point is represented by the stress element below. Determine the principal stresses and the maximum shear stresses, and draw the corresponding stress elements. Also, Analyse the stresses on an element inclined at 30° clockwise.



(4)

Step 1: Principal Stresses

$$\sigma_{\text{avg}} = \frac{-80 + 50}{2} = \frac{-30}{2} = -15 \text{ MPa}$$

$$R = \sqrt{\left(\frac{-80 - 50}{2}\right)^2 + 25^2} = \sqrt{\left(\frac{-130}{2}\right)^2 + 25^2} = \sqrt{(-65)^2 + 625} = \sqrt{4225 + 625} = \sqrt{}$$

$$\sigma_1 = \sigma_{\text{avg}} + R = -15 + 69.64 = 54.64 \text{ MPa}$$

$$\sigma_2 = \sigma_{\text{avg}} - R = -15 - 69.64 = -84.64 \text{ MPa}$$

Step 2: Maximum Shear Stress

$$\tau_{\text{max}} = R = 69.64 \text{ MPa}$$

Step 3: Stresses on element inclined at 30° clockwise ($\theta = 30^\circ$)

$$\sigma'_x = \sigma_{\text{avg}} + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy} \sin 2\theta$$

$$= -15 + \left(\frac{-80 - 50}{2}\right)\cos 60^\circ + 25 \sin 60^\circ$$

$$= -15 + (-65)(0.5) + 25(0.866) = -15 - 32.5 + 21.65 = -25.85 \text{ MPa}$$

$$\sigma_y' = \sigma_{\text{avg}} - \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta - \tau_{xy}\sin 2\theta$$

$$= -15 - (-65)(0.5) - 25(0.866) = -15 + 32.5 - 21.65 = -4.15 \text{ MPa}$$

$$\tau_{x'y'} = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$= -(-65)(0.866) + 25(0.5) = 56.29 + 12.5 = 68.79 \text{ MPa}$$

Parameter	Value (MPa)
Principal Stress σ_1	54.64
Principal Stress σ_2	-84.64
Maximum Shear Stress τ_{max}	69.64
σ_x' (30° inclined)	-25.85
σ_y' (30° inclined)	-4.15
$\tau_{x'y'}$ (30° inclined)	68.79